Exercises for Exam preparation

Exercise 1

Read the following scenario and answer the questions:

A new bakery opens in Algiers. The owner buys ovens, hires workers, purchases flour, and begins selling pastries to households. The government inspects the bakery to ensure food safety.

1.Identify the economic agents involved (households, firms, government, financial institutions).

2.Identify the four factors of production used (land, labor, capital, entrepreneurship) .

3.Explain whether the bakery contributes to microeconomics or macroeconomics analysis (and why).

Exercise 2

A bakery produces 200 baguettes per day with 10 workers. After introducing division of labor (mixing, shaping, baking, packaging), output increases to 320 baguettes per day.

1.Explain **why productivity increased**?

2.List **two possible drawbacks** the bakery may face due to highly repetitive tasks.

3.Suppose mechanization reduces the number of workers from 10 to 8 while producing 350 baguettes.

- Calculate productivity before and after the change.
- Interpret the impact on efficiency.

Exercise 3

Price (DA)	Qd	Qs
1,000	300	80
2,000	250	120
3,000	200	200
4,000	150	260

1. Determine the **equilibrium price and quantity** (where Qd = Qs) .
2. At a price of 1,000 DA, identify whether there is a surplus or shortage and calculate its size.
3. At a price of 4,000 DA, identify whether there is a surplus or shortage and calculate its size.
4. Explain what market forces will do in each case until equilibrium is restored.

Exercise 4

Suppose consumer income increases.

1. Explain whether demand shifts left or right for:

    1. Normal goods

    2. Inferior goods

2. Draw conclusions based on the demand shift causes.

Exercise 5

A sudden rise in the price of wheat (a key input for bread) occurs.

1. Does supply shift left or right?
2. What happens to equilibrium price and quantity?
3. Identify one real-life example of a supply shock in Algeria (weather, input cost, import restrictions, etc.).

Exercise 6

If the Price increases from **200 DA → 250 DA**.

And the Quantity demanded decreases from **500 units → 400 units**.

Using the midpoint formula from Lesson 5 & 6 :

1. Calculate % change in quantity.
2. Calculate % change in price.
3. Compute the price elasticity of demand.
4. State whether demand is elastic, inelastic, or unit elastic.
5. Explain what this means for a firm's revenue.

Exercise 7

A farmer supplies 600 kg of potatoes when price is 100 DA/kg.

When price rises to 130 DA/kg, he supplies 900 kg.

1. Use midpoint formula to compute elasticity of supply.
2. Classify supply: elastic or inelastic.
3. Explain why agricultural supply is often more inelastic in the short run

Exercise 9

A smartphone manufacturer faces the following situation:

•Due to division of labor and improved technology, productivity increases by 30% .

•The price of a key input (microchips) rises sharply.

•Consumers' income increases due to economic growth.

•The price elasticity of demand for this smartphone is elastic.

Questions:

1.Explain how improved productivity affects the firm's supply curve.

2.Explain how rising microchip prices affect the supply curve.

3.Describe how rising income affects demand for smartphones (assume they are a normal good).

4.Predict the new equilibrium price and quantity (direction only).

5.If the firm raises the price, what happens to revenue given demand is elastic?

6.Identify which economic agents are involved and how they interact.